

ELITE MATHEMATICS

Pearson Edexcel International GCSE

# Modular Unit 2

## Adaptive Classified

OL-T30 • Functions

### With Answers

Questions with full worked solutions

■ 50 pages

# FAMILY MAP

## Functions

|            |                                                |                 |
|------------|------------------------------------------------|-----------------|
| <b>F01</b> | <b>Interpret function notation and mapping</b> | 2 Q   2 marks   |
| <b>F02</b> | <b>Compose functions in the stated order</b>   | 4 Q   9 marks   |
| <b>F03</b> | <b>Control domain and range</b>                | 5 Q   11 marks  |
| <b>F04</b> | <b>Construct an inverse function</b>           | 12 Q   48 marks |

**16**  $f(x) = 9 - \sqrt{x}$       where  $x \geq 0$

$g(x) = 4x^2$

(a) Find  $f(9)$

(1)

**Worked Solution 09**

Function substitution

**UNIT 2****1 marks****Substitute**

$$f(9) = 9 - \sqrt{9} = 9 - 3 = 6.$$

**Final answer**

$$f(9) = 6$$

**16**  $f(x) = 9 - \sqrt{x}$       where  $x \geq 0$   
 $g(x) = 4x^2$

- (b) Solve  $fg(x) < 0$   
Show clear algebraic working.

.....  
(3)

**Worked Solution 10**

Composite function inequality

**UNIT 2**  
**3 marks****Compose the functions rigorously**

$$fg(x) = f(4x^2) = 9 - \sqrt{4x^2} = 9 - 2|x|.$$

**Solve the inequality as written**

$$9 - 2|x| < 0 \Rightarrow |x| > \frac{9}{2} \Rightarrow x < -\frac{9}{2} \text{ or } x > \frac{9}{2}.$$

**Read the domain statement carefully**

The printed  $x \geq 0$  is the input domain of  $f$ . Since  $g(x) = 4x^2 \geq 0$  for every real  $x$ , it does not remove the negative branch.

**If the original variable is additionally restricted**

If the original variable is also restricted to  $x \geq 0$ , then  $|x| = x$ , so the result reduces to  $x > \frac{9}{2}$ .

**Final answer**

$$\text{As written: } x < -\frac{9}{2} \text{ or } x > \frac{9}{2}.$$

$$\text{With an additional } x \geq 0 \text{ restriction: } x > \frac{9}{2}.$$

**Question 07**

4MA1-1H Q15

ORIGINAL QUESTION

4 marks

15 The function  $f$  is defined as

$$f: x \mapsto \frac{3x + 1}{x - 2}$$

(a) State the value of  $x$  that cannot be included in any domain of the function  $f$

.....  
(1)

(b) Express the inverse function  $f^{-1}$  in the form  $f^{-1}(x) = \dots$

$f^{-1}(x) = \dots$   
(3)

**Worked Solution 07**

Domain and inverse function

MODULAR UNIT 2

4 marks

**Domain**

$$x - 2 \neq 0 \Rightarrow x \neq 2.$$

**Inverse**

$$y = (3x + 1)/(x - 2) \Rightarrow x(y - 3) = 2y + 1 \Rightarrow f^{-1}(x) = \frac{2x + 1}{x - 3}.$$

**Final answer**

$$x \neq 2; \quad f^{-1}(x) = \frac{2x + 1}{x - 3}$$

**Question 31**

4MA1-2H Q19 (a)

ORIGINAL QUESTION UNIT 2

1 marks

19 The functions  $f$  and  $g$  are such that

$$f:x \mapsto 5x + 7$$

$$g:x \mapsto \frac{5}{2x-9}$$

(a) State which value of  $x$  cannot be included in any domain of  $g$

.....  
(1)

**Worked solution 31**

Excluded value from a rational function domain

MODULAR UNIT 2

1 marks

**(a) – Excluded value from a rational function domain**

Set the denominator to zero

$$2x - 9 = 0 \Rightarrow x = \frac{9}{2} = 4.5.$$

**Final answer** $x = 4.5$  cannot be included

**Question 32**

4MA1-2H Q19 (b)

ORIGINAL QUESTION UNIT 2

2 marks

19 The functions  $f$  and  $g$  are such that

$$f:x \mapsto 5x + 7$$

$$g:x \mapsto \frac{5}{2x-9}$$

(b) Find  $fg(4)$

.....  
(2)

**Worked solution 32**

Composite function value

MODULAR UNIT 2

2 marks

**(b) – Composite function value**Evaluate  $g(4)$ 

$$g(4) = \frac{5}{2(4) - 9} = -5.$$

Apply  $f$ 

$$fg(4) = f(-5) = 5(-5) + 7 = -18.$$

**Final answer**

-18

**Question 33**

4MA1-2H Q19 (c)

ORIGINAL QUESTION UNIT 2

4 marks

The function  $h$  is such that

$$h:x \mapsto 3x^2 - 12x + 8 \quad \text{where } x > 2$$

(c) Express the inverse function  $h^{-1}$  in the form  $h^{-1}:x \mapsto \dots$

$$h^{-1}:x \mapsto \dots \dots \dots$$

(4)

**Worked solution 33**

Find the inverse of a restricted quadratic

MODULAR UNIT 2

4 marks

**(c) – Find the inverse of a restricted quadratic**

Complete the square

$$y = 3x^2 - 12x + 8 = 3(x - 2)^2 - 4.$$

Rearrange

$$y + 4 = 3(x - 2)^2 \Rightarrow (x - 2)^2 = \frac{y + 4}{3}.$$

Use the restriction

$$x > 2 \Rightarrow x - 2 > 0, \text{ so take the positive square root.}$$

Swap variables

$$h^{-1}(x) = 2 + \sqrt{\frac{x + 4}{3}}.$$

**Final answer**

$$h^{-1}(x) = 2 + \sqrt{(x+4)/3}$$

**Question 14**

4MA1-1H Q17

ORIGINAL QUESTION

4 marks

17 The functions  $g$  and  $h$  are such that

$$g(x) = \frac{11}{2x - 5}$$

$$h(x) = x^2 + 4 \quad x \geq 0$$

(a) What value of  $x$  must be excluded from any domain of  $g$ ?

.....  
(1)

(b) Solve  $gh(x) = 1$

.....  
(3)

**Worked Solution 14**

Domain exclusion and composite equation

MODULAR UNIT 2

4 marks

**Exclude a zero denominator**

$$2x - 5 \neq 0 \Rightarrow x \neq \frac{5}{2}.$$

**Form the composite**

$$g(h(x)) = \frac{11}{2(x^2 + 4) - 5} = 1.$$

**Solve with the stated nonnegative domain**

$$2x^2 + 3 = 11 \Rightarrow x^2 = 4, \quad x = 2.$$

**Final answer**

$$x \neq \frac{5}{2}; \quad x = 2 \text{ for } g(h(x)) = 1$$

**Original question 17**

4MA1-1H Q25 M01

MODULAR UNIT 2

4 marks

25 The function  $f$  is such that  $f(x) = 2x^2 - 24x + 7$  where  $x \geq 6$

Find the inverse function  $f^{-1}(x)$

$$f^{-1}(x) = \dots\dots\dots$$

**Worked solution 17**

Find the inverse function

MODULAR UNIT 2

4 marks

**M01 – Find the inverse function****Complete the square**

$$f(x) = 2x^2 - 24x + 7 = 2(x - 6)^2 - 65.$$

**Rearrange**

$$y + 65 = 2(x - 6)^2 \Rightarrow (x - 6)^2 = (y + 65)/2.$$

**Use the domain  $x \geq 6$** 

$$x = 6 + \sqrt{(y + 65)/2}.$$

**Swap x and y**

$$f^{-1}(x) = 6 + \sqrt{(x + 65)/2}.$$

**Final answer**

$$f^{-1}(x) = 6 + \sqrt{\frac{x + 65}{2}}$$

**Original question 13**

4MA1-1H Q14 Q14(a)

**ORIGINAL QUESTION****1 marks****14** The function  $f$  is defined as

$$f: x \mapsto \frac{2x}{x-6} \quad x \neq 6$$

(a) Find  $f(10)$ 

---

**(1)**

**Worked solution 13**

Chapter: Functions

MODULAR UNIT 2

1 marks

**Q14(a) – Chapter: Functions**

Family: Interpret function notation and mapping • Variant: Evaluate a function at a numerical input

**Method**

$$f(10) = 2(10)/(10 - 6) = 20/4 = 5.$$

*Why: Substitute  $x = 10$  into the function definition.***Final answer****f(10): 5**

**Original question 14**

4MA1-1H Q14 Q14(b)

**ORIGINAL QUESTION****3 marks**

14 The function  $f$  is defined as

$$f: x \mapsto \frac{2x}{x-6} \quad x \neq 6$$

(b) Express the inverse function  $f^{-1}$  in the form  $f^{-1}: x \mapsto \dots$

$$f^{-1}: x \mapsto \dots \dots \dots (3)$$

**Worked solution 14**

Chapter: Functions

MODULAR UNIT 2

3 marks

**Q14(b) – Chapter: Functions**

Family: Construct an inverse function • Variant: Inverse of a linear or fractional function

**Method**Let  $y = 2x/(x - 6)$ ; then  $y(x - 6) = 2x$ .*Why: Introduce  $y$  and clear the denominator.***Method** $yx - 6y = 2x$ , so  $x(y - 2) = 6y$  and  $x = 6y/(y - 2)$ .*Why: Collect the terms containing  $x$  and factor them.***Method**Interchanging the input and output letters gives  $f^{-1}(x) = 6x/(x - 2)$ .*Why: The inverse reverses the mapping between  $x$  and  $y$ .***Final answer****inverse function:**  $f^{-1}(x) = 6x/(x - 2)$

**Original question 36**

4MA1-2H Q25 Q25(a)

**ORIGINAL QUESTION****4 marks**

25 The function  $g$  is defined as

$$g:x \mapsto 5 + 6x - x^2 \quad \text{with domain } \{x:x \geqslant 3\}$$

(a) Express the inverse function  $g^{-1}$  in the form  $g^{-1}:x \mapsto \dots$

$$g^{-1}:x \mapsto \dots \quad (4)$$

**Worked solution 36**

Chapter: Functions

MODULAR UNIT 2

4 marks

**Q25(a) – Chapter: Functions**

Family: Construct an inverse function • Variant: Inverse of a restricted quadratic function

**Method**

$$g(x) = 5 + 6x - x^2 = 14 - (x-3)^2.$$

*Why: Complete the square.***Method**For  $x \geq 3$ ,  $y = 14 - (x-3)^2$ , so  $(x-3)^2 = 14 - y$ .*Why: Use the restricted domain to choose the positive branch.***Method**

$$x-3 = \sqrt{(14-y)}.$$

*Why: Take the positive square root.***Method**Therefore  $g^{-1}(x) = 3 + \sqrt{(14-x)}$ .*Why: Swap  $x$  and  $y$  to write the inverse function.***Final answer**

inverse:  $g^{-1}(x) = 3 + \sqrt{(14-x)}$

**Original question 37**

4MA1-2H Q25 Q25(b)

**ORIGINAL QUESTION****1 marks**

(b) State the domain of  $g^{-1}$

(1)

**Worked solution 37**

Chapter: Functions

MODULAR UNIT 2

1 marks

**Q25(b) – Chapter: Functions**

Family: Control domain and range • Variant: Domain and range swap under inversion

**Method**

The square root requires  $14-x \geq 0$ , so the domain of  $g^{-1}$  is  $x \leq 14$ .

*Why: Use the radicand condition.*

**Final answer****domain:**  $x \leq 14$

**21** The functions  $f$  and  $g$  are such that

$$f(x) = x^2 - 2x \qquad g(x) = x + 3$$

The function  $h$  is such that  $h(x) = fg(x)$  for  $x \geq -2$

Express the inverse function  $h^{-1}(x)$  in the form  $h^{-1}(x) = \dots$

$$h^{-1}(x) = \dots\dots\dots$$

# Worked solution

January 2020 | Question 21 | 5 marks

Family: Construct an inverse function | Variant: Inverse of a composite-defined function

## METHOD / WORKING

1.  $g(x) = x + 3$ , so  $f(g(x)) = (x + 3)^2 - 2(x + 3) = x^2 + 4x + 3$ .

2. Complete square:  $h(x) = (x+2)^2 - 1$ , with  $x \geq -2$ .

3. Let  $y = (x+2)^2 - 1$ .

4. Then  $(x+2)^2 = y+1$  and  $x+2 = \sqrt{y+1}$  on the restricted domain.

5. Swap  $x, y$ :  $h^{-1}(x) = \sqrt{x+1} - 2$ .

## FINAL ANSWER

$$h^{-1}(x) = \sqrt{x+1} - 2$$

**22** The function  $f$  is such that  $f(x) = x^2 - 8x + 5$  where  $x \leq 4$

Express the inverse function  $f^{-1}$  in the form  $f^{-1}(x) = \dots$

$f^{-1}(x) = \dots\dots\dots$

## Worked solution

January 2021 | Question 22 | 3 marks

Family: Construct an inverse function • Variant: Inverse of a restricted quadratic function

Method / working

1. Complete the square:  $f(x) = (x-4)^2 - 11$  with  $x \leq 4$ .
  2. Set  $y = (x-4)^2 - 11$  and choose  $x-4 = -\sqrt{y+11}$  because  $x \leq 4$ .
  3. Swap  $x$  and  $y$ :  $f^{-1}(x) = 4 - \sqrt{x+11}$ .
- Final answer:  $f^{-1}(x) = 4 - \sqrt{x+11}$

24 The functions  $f$  and  $g$  are defined as

$$f(x) = 5x^2 - 10x + 7 \quad \text{where } x \geq 1$$

$$g(x) = 7x - 6$$

(a) Find  $fg(2)$

(2)

(b) Express the inverse function  $f^{-1}$  in the form  $f^{-1}(x) = \dots$

$f^{-1}(x) = \dots$   
(4)

## Worked solution

June 2021 | Question 24 | 2 marks

Family: Compose functions in the stated order • Variant: Evaluate a numerical composite function

Method / working

1.  $g(2) = 7 \times 2 - 6 = 8.$

2.  $f(g(2)) = f(8) = 5(8)^2 - 10(8) + 7 = 247.$

Final answer: 247

**24** The functions  $f$  and  $g$  are defined as

$$f(x) = 5x^2 - 10x + 7 \quad \text{where } x \geq 1$$

$$g(x) = 7x - 6$$

(a) Find  $fg(2)$

(2)

(b) Express the inverse function  $f^{-1}$  in the form  $f^{-1}(x) = \dots$

$f^{-1}(x) = \dots$   
(4)

# Worked solution

June 2021 | Question 24 | 4 marks

Family: Construct an inverse function | Variant: Inverse of a restricted quadratic function

## METHOD / WORKING

1. Write  $y = 5x^2 - 10x + 7 = 5(x - 1)^2 + 2$  with  $x \geq 1$ .
2. Rearrange:  $(x-1)^2 = (y-2)/5$ .
3. Use the domain  $x \geq 1$ , so  $x-1$  is the positive square root.
4. Swap  $x$  and  $y$ :  $f^{-1}(x) = 1 + \sqrt{(x-2)/5}$ .

## FINAL ANSWER

$$f^{-1}(x) = 1 + \sqrt{(x-2)/5}$$

15 The functions  $f$  and  $g$  are such that

$$f(x) = 2x - 3$$

$$g(x) = \frac{x}{3x + 1}$$

(a) State the value of  $x$  that cannot be included in any domain of  $g$

.....  
(1)

(b) Find  $gf(x)$   
Simplify your answer.

$gf(x) =$  .....  
(2)

(c) Express the inverse function  $g^{-1}$  in the form  $g^{-1}(x) = \dots$

$g^{-1}(x) =$  .....  
(3)

# Worked solution

November 2021 | Question 15 | 1 marks

Family: Control domain and range | Variant: Exclude a denominator zero from the domain

## METHOD / WORKING

$$1. f(x) = 2x - 3, g(x) = \frac{x}{3x + 1}$$

## FINAL ANSWER

Excluded value  $-\frac{1}{3}$ , gf

15 The functions  $f$  and  $g$  are such that

$$f(x) = 2x - 3$$

$$g(x) = \frac{x}{3x + 1}$$

(a) State the value of  $x$  that cannot be included in any domain of  $g$

.....  
(1)

(b) Find  $gf(x)$   
Simplify your answer.

$gf(x) =$  .....  
(2)

(c) Express the inverse function  $g^{-1}$  in the form  $g^{-1}(x) = \dots$

$g^{-1}(x) =$  .....  
(3)

# Worked solution

November 2021 | Question 15 | 2 marks

Family: Compose functions in the stated order | Variant: Derive a composite function rule

## METHOD / WORKING

1. (a) The denominator of  $g$  cannot be zero:

2.  $3x+1=0$

3.  $x=-\frac{1}{3}$

## FINAL ANSWER

$$(x) = \frac{2x-3}{6x-8}, \text{ and } g^{-1}$$

15 The functions  $f$  and  $g$  are such that

$$f(x) = 2x - 3$$

$$g(x) = \frac{x}{3x + 1}$$

(a) State the value of  $x$  that cannot be included in any domain of  $g$

.....  
(1)

(b) Find  $gf(x)$   
Simplify your answer.

$gf(x) =$  .....  
(2)

(c) Express the inverse function  $g^{-1}$  in the form  $g^{-1}(x) = \dots$

$g^{-1}(x) =$  .....  
(3)

# Worked solution

November 2021 | Question 15 | 3 marks

Family: Construct an inverse function | Variant: Inverse of a linear or fractional function

## METHOD / WORKING

1. (b)

2.  $gf(x)=g(f(x))$

3.  $gf(x) = \frac{2x-3}{3(2x-3)+1}$

4.  $gf(x) = \frac{2x-3}{6x-8}$

## FINAL ANSWER

$$(x) = \frac{x}{1-3x}$$

21 The function  $f$  is such that  $f(x) = 5 + 6x - x^2$  for  $x \leq 3$

(a) Express  $5 + 6x - x^2$  in the form  $p - (x - q)^2$  where  $p$  and  $q$  are constants.

.....  
(2)

(b) Using your answer to part (a), find the range of values of  $x$  for which  $f^{-1}(x)$  is positive.

.....  
(5)

.....  
(Total for Question 21 is 7 marks)

## Answer reference | May-Nov 2020 | Question 21

Family: Construct an inverse function • Variant: Inverse of a restricted quadratic function

|               |   |                                                                                                                                                |                  |   |                                                                                                                                                                            |
|---------------|---|------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 21            | a | $5 - (x \pm q)^2 + 9$ oe <b>or</b> $p - (x - 3)^2$ oe<br><br><b>or</b><br>$p - q^2 + 2qx - x^2$ and one of<br>$2q = 6$ <b>or</b> $p - q^2 = 5$ |                  |   | M1 may be seen in working eg $-(x - 3)^2 - 9 - 5]$<br><br><b>or</b><br>expanding $p - (x - q)^2$ correctly and equating one of the coefficient of $x$ or the constant term |
|               |   |                                                                                                                                                | $14 - (x - 3)^2$ | 2 | A1 fully correct<br><br>SCB1 for $(x - 3)^2 - 14$                                                                                                                          |
|               | b | e.g. $(x - 3)^2 = 14 - y$<br>[or $(y - 3)^2 = 14 - x]$                                                                                         |                  |   | M1 correct steps to isolate their bracket<br>fit from (a) dep on expression in form $\pm p \pm (x - q)^2$                                                                  |
|               |   | $x = 3 \pm \sqrt{14 - y}$<br>[or $y = 3 \pm \sqrt{14 - x}$ ]                                                                                   |                  |   | M1 complete method to find $y$ in terms of $x$ or $x$ in terms of $y$ . Condone + for $\pm$<br>fit from (a) dep on expression in form $\pm p \pm (x - q)^2$                |
|               |   | $(f^{-1}(x) =) 3 - \sqrt{14 - x}$                                                                                                              |                  |   | M1 for the correct inverse                                                                                                                                                 |
|               |   |                                                                                                                                                |                  |   | M1 method to solve $0 < 3 - \sqrt{14 - x}$ or a lower bound of 5 clearly shown, eg $x > 5$ as part of the answer                                                           |
|               |   |                                                                                                                                                | $5 < x \leq 14$  | 5 | A1<br>cao                                                                                                                                                                  |
| Total 7 marks |   |                                                                                                                                                |                  |   |                                                                                                                                                                            |

## Worked solution

May-Nov 2020 | Question 21 | 7 marks

Family: Construct an inverse function • Variant: Inverse of a restricted quadratic function

Official final mark-scheme pages are embedded immediately after this question.

18 The function  $f$  is such that  $f(x) = \frac{k}{x}$  where  $x \neq 0$  and  $k$  is an integer.

(a) Express the inverse function  $f^{-1}$  in the form  $f^{-1}(x) = \dots$

$$f^{-1}(x) = \dots \quad (1)$$

The function  $g$  is such that  $g(x) = 2 - 3x^4$  where  $x \neq 0$

The function  $h$  is such that  $h(x) = \frac{3x}{2-x}$  where  $x \neq 2$

(b) (i) Find  $g(-2)$

$$\dots \quad (1)$$

(ii) Express the composite function  $hg$  in the form  $hg(x) = \dots$   
Give your answer in its simplest form.

$$hg(x) = \dots \quad (2)$$

(Total for Question 18 is 4 marks)

## Answer reference | January 2023 | Question 18

Family: Construct an inverse function • Variant: Inverse of a composite-defined function

|    |        |                                                                                            |                      |   |                                 |
|----|--------|--------------------------------------------------------------------------------------------|----------------------|---|---------------------------------|
| 18 | (a)    |                                                                                            | $\frac{k}{x}$        | 1 | B1 allow $kx^{-1}$              |
|    | (b)(i) |                                                                                            | -46                  | 1 | B1 cao                          |
|    | (ii)   | $\frac{3(2-3x^4)}{2-(2-3x^4)}$ or $\frac{6-9x^4}{2-2+3x^4}$ oe or $\frac{6-9x^4}{3x^4}$ oe |                      | 2 | M1                              |
|    |        | <i>Correct answer scores full marks (unless from obvious incorrect working)</i>            | $\frac{2-3x^4}{x^4}$ |   | A1 allow $\frac{2}{x^4} - 3$ oe |
|    |        |                                                                                            |                      |   | <b>Total 4 marks</b>            |

## Worked solution

January 2023 | Question 18 | 4 marks

Family: Construct an inverse function • Variant: Inverse of a composite-defined function

Official final mark-scheme pages are embedded immediately after this question.

25 The function  $f$  is such that  $f(x) = 3x^2 - 12x + 7$  where  $x \leq 2$

Express the inverse function  $f^{-1}$  in the form  $f^{-1}(x) = \dots$

$f^{-1}(x) = \dots\dots\dots$

(Total for Question 25 is 4 marks)

## Answer reference | January 2023 | Question 25

Family: Construct an inverse function • Variant: Inverse of a restricted quadratic function

| Q      | working                                                                                                                      | Answer                     | Mark | Notes                                                                                                    |
|--------|------------------------------------------------------------------------------------------------------------------------------|----------------------------|------|----------------------------------------------------------------------------------------------------------|
| 25     | $y = 3(x^2 - 4x) + 7$ or $y = 3\left(x^2 - 4x + \frac{7}{3}\right)$ or $\frac{y-7}{3} = x^2 - 4x$<br>or $y = 3(x-2)^2 \dots$ |                            | 4    | M1<br>for a correct equation for a first step to complete the square                                     |
|        | eg $y = 3((x-2)^2 - 2^2) + 7$ or $y = 3((x-2)^2 - 2^2 + \frac{7}{3})$ or<br>$y = 3(x-2)^2 - 5$ oe or                         |                            |      | M1                                                                                                       |
|        | $(x-2)^2 = \frac{y+5}{3}$ oe eg $(x-2)^2 = \frac{y-7}{3} + 4$ or<br>$x-2 = (\pm)\sqrt{\frac{y+5}{3}}$ oe                     |                            |      | M1                                                                                                       |
|        | Correct answer scores full marks (unless from obvious incorrect working)                                                     | $2 - \sqrt{\frac{x+5}{3}}$ |      | A1<br>oe<br>NB: note only negative square root.<br>Must be in terms of $x$<br><b>any equivalent form</b> |
|        | <b>Note: Allow candidates to swap <math>x</math> and <math>y</math> when finding the inverse</b>                             |                            |      | <b>Total 4 marks</b>                                                                                     |
| 25 Alt | $3x^2 - 12x + (7 - y) = 0$                                                                                                   |                            | 4    | M1<br>for a correct first step                                                                           |
|        | $(x =) \frac{12 \pm \sqrt{144 - 12(7 - y)}}{6}$                                                                              |                            |      | M1                                                                                                       |
|        | $(x =) 2 \pm \sqrt{\frac{60 + 12y}{36}}$ oe                                                                                  |                            |      | M1                                                                                                       |
|        | Correct answer scores full marks (unless from obvious incorrect working)                                                     | $2 - \sqrt{\frac{x+5}{3}}$ |      | A1<br>oe<br>NB: note only negative square root.<br>Must be in terms of $x$<br><b>any equivalent form</b> |
|        |                                                                                                                              |                            |      | <b>Total 4 marks</b>                                                                                     |

## Worked solution

January 2023 | Question 25 | 4 marks

Family: Construct an inverse function • Variant: Inverse of a restricted quadratic function

Official final mark-scheme pages are embedded immediately after this question.

19 The functions  $f$  and  $g$  are such that

$$f(x) = 3x - 2$$

$$g(x) = \frac{x}{2x - 1}$$

(a) Find  $g(3)$

.....  
(1)

(b) Find  $gf(x)$

Give your answer in its simplest form.

$$gf(x) = \text{.....}$$

(2)

(Total for Question 19 is 3 marks)

# Worked solution

November 2025 | Question 19 | 3 marks

Family: Construct an inverse function | Variant: Inverse of a linear or fractional function

## METHOD / WORKING

|                                                                                              |
|----------------------------------------------------------------------------------------------|
| Worked solution                                                                              |
| November 2025   Question 19   3 marks                                                        |
| Family: Construct an inverse function   Variant: Inverse of a linear or fractional function  |
| Solution                                                                                     |
| $f(x)=3x-2, \text{quad } g(x)=(x)/(x-1)$                                                     |
| (a)                                                                                          |
| $g(3)=(3)/(3-1)=\frac{3}{2}$                                                                 |
| (b)                                                                                          |
| $gf(x)=g(f(x))$                                                                              |
| $gf(x)=(3x-2)/((3x-2)-1)$                                                                    |
| $gf(x)=(3x-2)/(3x-3)$                                                                        |
| <i>Answer : <math>g(3) = \frac{3}{2}</math>, and <math>gf(x) = \frac{3x-2}{3x-3}</math>.</i> |
| PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION                                |

## FINAL ANSWER